

Soft failure detection and identification in optical networks using cascaded deep learning model

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ABSTRACT

Due to malfunction of network devices and surge in physical layer impairments, the quality of transmission (QoT) in backbone optical networks may degrade. If the cause of the degradation is not timely diagnosed and addressed adequately, it may deteriorate into a hard failure. In this study, we consider the external cavity laser (ECL) malfunction-, erbium-doped fiber amplifier (EDFA) malfunction-, and nonlinear interference-related soft failures. We propose a software-defined optical network (SDON)-based soft failure detection and identification strategy using a cascaded deep learning model. Time-series QoT data of normal and degraded lightpaths obtained through the optical performance monitoring equipment is used to train the proposed cascaded deep learning model. In the first stage, a long short-term memory-based autoencoder (LSTM-AE) model is used as a binary classifier to identify the anomalous time-series sequences. Subsequently, an LSTM-based multiclass classifier is used to identify the type of soft failure. Our proposed approach shows an accuracy of 99.70%.

1. Introduction

Given the huge data rate supported by backbone optical networks, it is crucial to focus on controlling network device failures and improving network reliability. A network device failure can cause massive data loss and a long failure recovery cycle, resulting in a critical disruption of network services and a formidable loss of revenue. To avoid complete (hard) network failure, automatically recognizing soft failure, which may develop into hard failure becomes very important [1,2]. During the event of soft failure, the quality of transmission (QoT) parameters gradually degrade over time, even if the complete disruption of the received optical signal may not occur. Soft failures heavily affect network performance and may eventually turn into hard failures, causing complete disruption of network services. Therefore, determining the occurrence of soft failure (i.e., detection), identifying the cause of soft failure (i.e., identification), and finding the faulty device/network segment (i.e., localization) in an optical backbone network are highly desirable to prevent network failure scenarios [3,4].

With the recent advancement of optical performance monitoring (OPM) technologies [5], the collection and assessment of various critical QoT parameters and devices' characteristic parameters can be realized. In recent times, for intelligent and automatic network control, the software-defined networking (SDN) model is being used that separates the data plane and the control plane and consolidates network control within an SDN controller. SDN integrated optical network

(i.e., SDON) [6] implemented using OPM technologies receives up-to-date information about the current state of the network, and make necessary adjustments to the lightpaths, including the signal power levels, data rates, modulation formats, and spectrum assignment to improve the network operational performance [5,7].

The soft failure in backbone optical networks operating at extremely high data rates over long transmission distances is heavily influenced by factors, such as the nonlinear interference (NLI) developed in the optical fiber, deterioration of the erbium-doped fiber amplifier (EDFA) gain, and the laser malfunction at the transmitter. An increase in the lightpath's transmit power or the number of lightpaths accommodated in a fiber may increase the NLI power, leading to degradation in the QoT parameters. Any decrease in EDFA gain can decrease the received optical power level and degrade the QoT parameters of the lightpaths passing through the EDFA. Moreover, the performance of a transmitter laser can deteriorate as a result of the degradation of multiple internal or external parameters, such as the bias current and temperature. The aforementioned degradation causes the threshold current for transmitter laser to increase [8,9].

In general, traditional techniques for soft failure management rely on mathematical models and/or threshold limits, which may not be very accurate. Thus, the analytical approaches may not be very efficient [10,11]. Fortunately, rapidly advancing machine learning (ML) techniques provide a possible solution to this problem, with several

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relevant studies already published in the literature. In [12], random forest (RF), binary and multiclass support vector machine (SVM) and neural network (NN) algorithms are applied for the detection and identification of soft failures that may occur due to filter misalignment and amplifier gain reduction. In [13], an anomaly detection model based on SVM algorithm is proposed to detect anomalies in the signal-to-noise ratio (SNR) data. In [14], Ruiz et al. study the effects on QoT parameters (viz., received power and pre-forward error corrected (FEC) bit error rate (BER)) for detection of failures due to tight filtering and inter-channel interference. The authors also propose a failure localization algorithm based on Bayesian network (BN). In [15], a framework is proposed to improve the accuracy of the monitoring data and localize the soft failures. The authors consider soft failures due to the malfunction of amplifiers, tunable filters, transmitter, optical switches, and fiber aging and bending. The monitoring data including the dispersion, optical signal-to-noise ratio (OSNR), and amplified spontaneous emission (ASE) noise values, obtained from digital signal processing (DSP) aided coherent receivers, are analyzed using network kriging algorithm to localize soft failures at link level. In [16], a convolutional neural network (CNN)-based three-stage soft failure detection, identification, and lifetime prediction framework concerning EDFAs is proposed. The incidence of failures in backbone optical networks is minimal, as networks are often designed to be conservative, keeping into consideration significant operating margins [17,18]. Thus, it is challenging to obtain a failure dataset that is large enough to train the ML models. To correctly detect and identify soft failures, a synthetic training dataset is needed to be generated, thereby necessitating the optical network to be modeled in a close-to real-world fashion.

The literature discussed above does not exhaustively explore time correlations for the time series OPM data. Time series data analysis allows for identifying patterns and trends within data over time, helping to understand the underlying structures and behaviors, further facilitating detection of anomalies in the data. In the following, we discuss the literature that exploits time correlation for further enhancement of prognostic methods. Considering the time-series correlation, the authors in [19] propose a soft failure detection and localization scheme built upon a bi-directional gated recurrent unit (BiGRU) for effective time-series processing. The authors use a real-world monitoring dataset comprising input optical power, laser bias current, output optical power, etc., to capture temporal features from multiple optical transport network equipment. Even though accuracy in soft failure detection and identification is important, misclassification of anomalous data as normal should be critically avoided. Therefore, a cascaded approach, in which a two-stage classification is used, with the first stage used for detecting the soft failure and the second stage used for identifying the type of soft failure, could be suitable for soft failure mitigation. In a similar line, the cascaded approach is used in other areas of knowledge. Concerning denial-of-service (DoS) attack detection and classification in network-on-chip, a cascaded ML-based approach is used, wherein first, SVM is used for attack detection, and the k-nearest neighbor (KNN) model is used to further classify suspicious data to identify different types of attacks [20]. In [21], concerning different types of faults, namely fiber cut, fiber eavesdropping, dirty connector, and bad splice in a passive optical network environment, a gated recurrent unit (GRU)-based autoencoder (AE) model is used for anomaly detection. An attention-based BiGRU model is used for the identification and localization of fiber defects. In [8], a GRU model with an attention mechanism is employed to forecast the deterioration of laser performance, specifically the rise in laser current.

In this paper, we propose an SDON-based framework, in which we employ a cascaded soft failure detection and identification approach by combining the strengths of long short-term memory (LSTM) and AE. SDON network collects the QoT values of the lightpaths from the network nodes at a specific time interval. In the training phase, first, we train an unsupervised LSTM-based AE (LSTM-AE) model using only the normally operating lightpaths' QoT values, which are expected to

be available in abundance in a real network infrastructure. Thereafter, we train a supervised LSTM-based classifier model using anomalous lightpaths' (suffering from different types of soft failure) QoT values. In the testing phase, first, we use the LSTM-AE model to identify aberrant time-series sequences. Thereafter, we identify the type of soft failure using the LSTM-based classifier.

The following are the primary contributions of our proposed approach.

- To the best of our knowledge, for the first time, we collectively take into account the transmitter laser-, EDFA-, and NLI-related soft failures, which are considered to be the predominant sources of failures in a backbone optical network.
- We propose a novel deep learning-based cascaded approach for soft failure detection and identification in SDON framework. LSTM-AE model detects the anomalous QoT data sequences, which are subsequently passed through an LSTM classifier to identify the type of soft failure.
- Through exhaustive simulation, we show the relative performance of our proposed cascaded model compared to different ML (viz., LSTM, GRU, RF, and SVM) based classifiers.
- As the availability of real network data is scarce, we generate synthetic dataset for the training and testing of the deep learning models considering real-world backbone optical network settings, taking into account intricate details of node architecture and optical devices' characteristics.

The rest of the paper is organized as follows. In Section 2, we discuss the formulae to compute QoT for lightpaths and the device degradation characteristics. In Section 3, we propose an SDON-based framework for soft failure detection and identification. In Section 4, we provide the simulation setting and the dataset generation methodology. In Section 5, we discuss the results for our proposed cascaded deep learning model. Section 6 concludes the paper.

2. Estimation of device degradation characteristics and QoT of lightpaths

In this section, first we discuss the formulae to obtain the device degradation characteristics. We also compute ASE power, NLI power, OSNR, SNR, and pre-FEC BER at the receiver of a lightpath. The optical output power from a semiconductor laser (P_{out}) of a transmitter can be presented as [22]

$$P_{out} = \eta_{sl}(I - I_{th}), \quad (1)$$

where I is the operating current, I_{th} is the threshold current, and η_{sl} is the slope efficiency. In the event of semiconductor laser failure, the degradation process occurs in the active region, leading to an increase in the internal absorption loss and shortening the injected carrier lifetime (due to an increase in the nonradiative recombination). Consequently, the slope efficiency may fall and (or) the threshold current may rise, leading to a decrease in the optical output power. In order to maintain a constant transmit power P_{out} , the operating current is varied following the relation given below [23]

$$I(t) = I_0 + \beta \exp\left(P_{out}^q \exp\left(\mu_0 - \frac{E_a}{k_B T}\right)t\right), \quad (2)$$

where I_0 is the initial operating current, β is the nonradiative current, q is the derating exponent, μ_0 is the scale parameter, E_a is the activation energy, k_B is Boltzmann constant, T is the temperature (in absolute scale), and t is the operational time.

A malfunction in the pump laser of an EDFA may decrease the pump laser power causing degradation in the gain of the amplifier. Thus, the received optical power at the receiver of a lightpath traversed through the EDFA reduces. Under small signal approximation, the evolution of

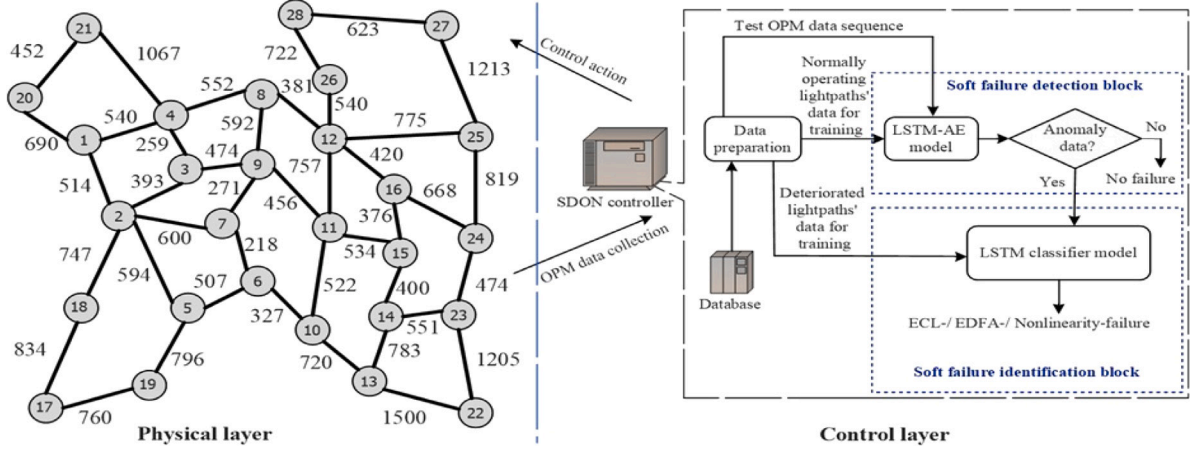


Fig. 1. SDON-based network architecture for soft failure detection and identification.

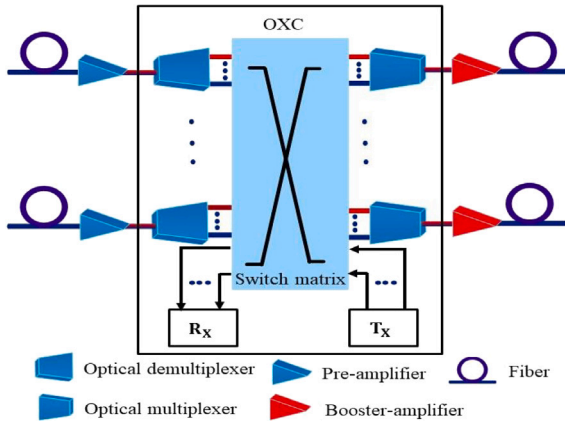


Fig. 2. General architecture of a network node [25,26].

the pump and signal powers along an erbium-doped fiber is presented by the following relations [24]:

$$\frac{dP_p}{dz} = \sigma_{pa} P_{p0} N_0 \ln \left[\frac{1 + \frac{P_p(z)}{P_{p0}} \exp(-b^2/\Omega_p^2)}{1 + \frac{P_p(z)}{P_{p0}}} \right], \quad (3)$$

$$\begin{aligned} \frac{dP_s}{dz} = & \sigma_{sa} N_0 P_s \frac{P_{p0}}{P_p} \left[\frac{\eta_s P_p}{P_{p0}} \left(1 - \exp(-b^2/\Omega_s^2) \right) \right. \\ & \left. + (1 + \eta_s) \ln \left(\frac{1 + \frac{P_p}{P_{p0}} \exp(-b^2/\Omega_s^2)}{1 + \frac{P_p}{P_{p0}}} \right) \right], \end{aligned} \quad (4)$$

where P_p and P_s denote the pump laser power and the signal power at a length z in the doped fiber, respectively. σ_{pa} is the pump absorption cross-section, N_0 is the erbium ion concentration, P_{p0} is the pump input power, and b is the doping radius. Ω_p (Ω_s) is a parameter that depends on the characteristics of the fiber and the pump (signal) wavelength. $\eta_s = \frac{\sigma_{se}}{\sigma_{sa}}$, where σ_{se} and σ_{sa} are the signal emission cross-section and signal absorption cross-section, respectively. The pump and signal powers are obtained by solving Eq. (3) and Eq. (4) using Runge-Kutta method.

Two major noise sources in optical networks are ASE and NLI, which degrade the $OSNR$ of a lightpath set up in the network. The $OSNR$ of a lightpath is computed as

$$OSNR = \frac{P_{Sig}}{P_{ASE} + P_{NLI}}, \quad (5)$$

where P_{Sig} , P_{ASE} , and P_{NLI} are the signal, ASE noise, and NLI noise powers at the end of the lightpath, respectively. The relation between $OSNR$ and SNR at the end of the lightpath is given as

$$SNR = \frac{B_0}{R_s} OSNR, \quad (6)$$

where B_0 is the noise bandwidth and R_s is the symbol rate. The pre-FEC BER for polarization-modulated quadrature phase shift keying (PM-QPSK) modulation is given by the relation

$$BER = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{SNR}{2}} \right). \quad (7)$$

We consider the backbone network and the node architecture as shown in Fig. 1 and Fig. 2, respectively, where inline amplifiers are placed on the fiber links at a regular interval (span). A node consists of an OXC, and multiple preamplifiers and booster amplifiers. The OXC comprises an optical switch matrix and multiple optical demultiplexers, optical multiplexers, and transceivers allowing for the add and drop of multiple lightpaths at the node via the switch matrix. A lightpath generated from a transmitter of a node is attenuated due to switch loss and multiplexer loss. The loss is compensated by the booster amplifier within the node. On the other hand, the optical power of a lightpath terminated at a receiver of a node is attenuated due to loss in the fiber segment (between the last inline EDFA traversed and the given node) and the demultiplexer. The preamplifier compensates for this attenuation. Additionally, the fiber loss in each span is compensated by the gain of the accompanying inline amplifier. We consider that a lightpath traversing through several nodes and fiber links, in general, consists of N_{In} number of inline amplifiers, N_p number of preamplifiers, and N_B number of booster amplifiers. The ASE noise power at the receiver can be computed as [25–27]

$$\begin{aligned} P_{ASE} = & N_{In} F_{edfa} h\nu (G_{In} - 1) B_0 + \\ & N_p F_{edfa} h\nu (G_p - 1) B_0 + N_B F_{edfa} h\nu (G_B - 1) B_0, \end{aligned} \quad (8)$$

where F_{edfa} is the noise figure of EDFAs, h is Planck's constant, ν is the central operating frequency, and G_{In} , G_p , and G_B are the gain of the inline amplifiers, preamplifiers, and booster amplifiers, respectively.

The nonlinear noise power at the receiver for a given lightpath using the incoherent Gaussian noise model is presented as [28]

$$P_{NLI} = \frac{8}{27} N_S \gamma^2 L_{eff}^2 P_T^3 \frac{\ln(\pi^2 \beta L_{eff,a} N_{Ch}^2 R_s^2)}{\pi \beta L_{eff,a} R_s^3} B_0, \quad (9)$$

where N_S is the total number of spans in the fiber links, γ is the nonlinear coefficient of fiber, P_T is the transmit optical power of the lightpath, β is the dispersion parameter, and N_{Ch} is the number of lightpaths in each link. The effective fiber length (L_{eff}) is defined as $L_{eff} = \frac{1 - \exp(-2\alpha L_S)}{2\alpha}$, where L_S is the span length and α is the fiber

loss coefficient. The asymptotic effective length ($L_{eff,a}$) is defined as $L_{eff,a} = \frac{1}{2\alpha}$.

3. Proposed SDON-based framework for soft failure detection and identification

Fig. 1 illustrates the proposed SDON-based system for soft failure detection and identification. Each lightpath terminating at a node in the backbone optical network is equipped with an OPM facility. We consider that different QoT parameters, such as the received optical power, transmitter laser operating current, OSNR, SNR, and pre-FEC BER of each lightpath terminating at a node, are recorded at specific time intervals. ASE and NLI impairments are considered additive per link. We consider only three soft failure scenarios, viz., transmitter laser failure, inline amplifier gain-related failure, and NLI-related failure. We consider that at any given time, multiple lightpaths in the network can suffer from the same or different types of soft failures; however, more than one failure cannot simultaneously develop in any given lightpath. We employ the SDON controller that oversees the physical layer devices using the network control protocol. Each OPM at the nodes is tasked with providing the controller with the physical layer performance parameters. Our proposed cascaded deep learning model has two stages, viz., soft failure detection and soft failure identification. In the first stage, we employ a soft failure detection strategy based on LSTM-AE model, whereas, in the second stage, we execute soft failure identification using LSTM-based classifier model. As shown in Fig. 1, the SDON controller collects each node's OPM time-series data. The OPM data are stored in a network database and preprocessed for the training purposes of the LSTM-based deep learning models. Normally operating lightpaths' time sample points are used to train the LSTM-AE model, whereas the failure lightpaths' time sample points are used to train the LSTM-based classifier model. Subsequently, in the testing phase, when a new time-series QoT data for a lightpath arrives, the LSTM-AE model predicts whether operation of the given lightpath is normal or anomalous (i.e., suffering from soft failure). The anomalous time series data after being identified by the soft failure detection block is further passed through the soft failure identification block to identify the type of soft failure.

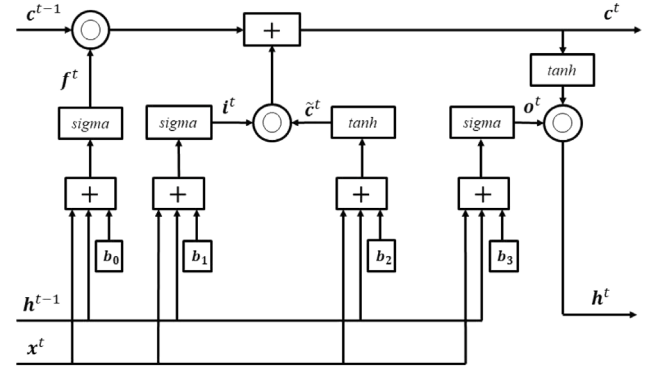
LSTMs can efficiently handle data sequences, making them robust for tasks where the input order and temporal dependencies of data are crucial. The model learns the time series dependency of lightpaths' QoT data sequence points and figures out the underlying patterns in the multivariate time series data for different soft failures and normal lightpath operations. LSTM can automatically learn relevant features from time series data, eliminating the need for manual feature engineering, and allowing the model to adapt to the complexity of the data.

Fig. 3 illustrates the structure of an *LSTM cell*. A single LSTM cell consists of three gates, viz., the input gate, forget gate, and output gate (denoted as i , f , and o , respectively), and weight matrix. Gates control the flow of information to the cell state (viz., c). Gates are represented as sigmoid activation functions (σ). The weights between the input (x^t) to the hidden layer (h^t) are represented as matrix U and the recurrent connection weights between the previous hidden layer (i.e., h^{t-1}) and the current layer (i.e., h^t) are represented as matrix W . The Hadamard product is represented by $\odot$. In an LSTM, four activation vectors correspond to the three gates and a cell input activation ($\tilde{c}$). The equations that describe the functionality of the LSTM gates are presented in the following [29]. The forget gate activation vector determines the percentage of long-term memory to be remembered,

$$f^t = \sigma(U^f x^t + W^f h^{t-1} + b_0). \quad (10)$$

The input gate activation vector controls the flow of new information into the cell state,

$$i^t = \sigma(U^i x^t + W^i h^{t-1} + b_1). \quad (11)$$



c^{t-1} : memory from the previous block, c^t : memory from current block, h^{t-1} : output of previous block, h^t : output of current block, x^t : input vector, b_i : bias ($i = 0, 1, 2, 3$), σ : sigmoid activation function, $\tanh$: hyperbolic tangent activation function, $\odot$: Hadamard product.

Fig. 3. Single LSTM cell structure.

The output gate activation vector determines which parts of the current cell state should be exposed to the next layer,

$$o^t = \sigma(U^o x^t + W^o h^{t-1} + b_3). \quad (12)$$

The cell input activation vector represents potential long-term memory by proposing new candidate values to be added to the cell state,

$$\tilde{c}^t = \tanh(U^{\tilde{c}} x^t + W^{\tilde{c}} h^{t-1} + b_2). \quad (13)$$

The cell state c^t is computed through the combination of the forget gate f^t operation with the old cell state c^{t-1} and the input gate i^t operation with the cell input activation vector,

$$c^t = f^t \odot c^{t-1} + i^t \odot \tilde{c}^t. \quad (14)$$

The new hidden state h^t is the output of the recurrent unit at a given timestamp

$$h^t = o^t \odot \tanh(c^t). \quad (15)$$

An AE is an unsupervised neural network that is trained to disregard inconsequential data to acquire a representation for a given collection of unlabeled input data. The process involves reducing the dimensionality of the input data and afterwards reconstructing the initial data from this transformed representation. The model employs a non-linear activation function and incorporates numerous layers to effectively capture and comprehend the intricate non-linear relationships present within the dataset. AE consists of two distinct components, namely the encoder and the decoder (Fig. 4). The encoder is trained to decrease the dimensions of the input vector x in accordance with the equation,

$$z = \phi(Wx + b) \quad (16)$$

where z represents the latent dimension, ϕ is the activation function, W is the weight matrix, and b is the bias. The decoder is also trained to obtain the reconstructed data as

$$\tilde{x} = \phi'(W'z + b'), \quad (17)$$

where ϕ' is the activation function, W' is the weight matrix, and b' is the bias. The primary objective of AE is to minimize the reconstruction error between the output $\tilde{x}$ and the original input x , thereby ensuring their similarity. The loss function can be expressed in terms of the mean absolute error (MAE) metric as

$$MAE = \frac{1}{l} \sum_{j=1}^l |x_j - \tilde{x}_j|, \quad (18)$$

where l is the length of the vector.

LSTM-AE [30] combines the strength of AE with LSTM cells, where the encoder and decoder are constructed using multiple LSTM cells (Fig.

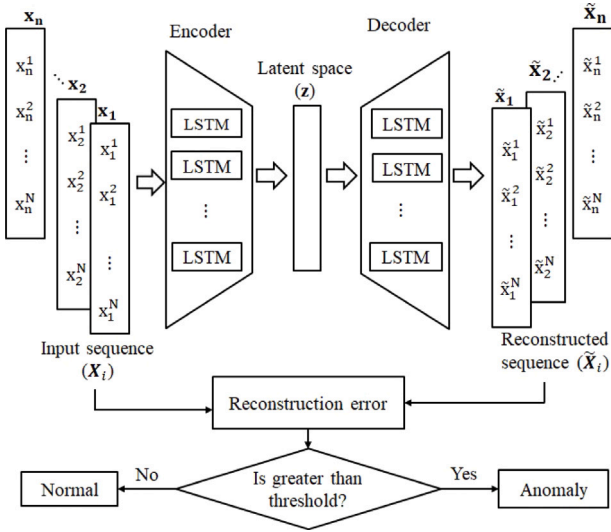


Fig. 4. LSTM-AE model for anomaly detection.

4). The input dataset having time series data, consists of N features and total M timesteps. Data at each timestep i is represented as $\mathbf{x}_i \in \mathbf{R}^{1 \times N}$, where $i \in \{1, 2, \dots, M\}$ and a sequence consisting of n time length window is represented as $\mathbf{X}_i = [\mathbf{x}_{i-n+1}, \mathbf{x}_{i-n+2}, \dots, \mathbf{x}_i]$, where $i > n$. The LSTM encoder section captures the salient features and dependencies in the multivariate time series sample in a condensed form. The LSTM decoder sequentially unfolds the input data to obtain the reconstructed sequence, restoring its original sequence structure after undergoing sequence folding on timesteps. The reconstruction error is computed following Eq. (18). A threshold (i.e., the maximum value of the reconstruction error) is established, and observations that surpass the threshold are considered anomalous.

We also construct an *LSTM classifier* by combining multiple LSTM layers followed by optional additional layers, such as dropout and dense layers. The LSTM layers are responsible for learning to extract the relevant features from the input sequences and capture long-range dependencies within the data. LSTM cells within each layer have internal mechanisms, such as gates, that allow them to selectively store or discard information over time, making them well suited for modeling sequential data. We train the LSTM classifier to learn the hidden patterns in the failure lightpaths' QoT data sequences, thereby identifying the type of soft failure.

4. Simulation setting and dataset generation

In this section, we describe the simulation setting, followed by a description of the dataset generation methodology.

4.1. Simulation setting

In this study, we consider a 28-node network with 41 links [31] (Fig. 1). Each node in the network is equipped with a coherent receiver that provides OPM facility to monitor QoT parameters, viz., noise (ASE/NLI) power, OSNR, SNR, and pre-FEC BER of the lightpaths terminating at the node. The nodes are interconnected by bidirectional links, wherein each link comprises two standard single-mode unidirectional fibers that support traffic flow in opposite directions. We employ standard single-mode fiber (SSMF) in the network with the physical layer simulation parameters as shown in Table 1. The fiber has an attenuation coefficient of 0.22 dB/km, a nonlinear coefficient of 1.3 /W/km, and a dispersion parameter of 16.7 ps/nm/km. Inline amplifiers are placed at a regular interval (i.e., span) of 80 km. Considering the node architecture as shown in Fig. 2, we assume a fixed loss of 2 dB for each multiplexer,

Table 1
Physical layer simulation parameters [28].

Parameter name	Value
Fiber attenuation coefficient	0.22 dB/km
Fiber nonlinear coefficient	1.3 /W/km
Fiber dispersion parameter	16.7 ps/nm/km
Symbol rate	32 GBaud
Noise figure of EDFA	5 dB
Span length	80 Km
Central operating frequency	193.1 THz
Multiplexer loss	2 dB
Demultiplexer loss	2 dB
Switch loss	2 dB
Noise bandwidth, B_0	12.5 GHz

Table 2
ECL parameters during normal operation [32].

Parameter name	Value
Threshold current	26.7 mA
Transmit power	-2 dBm
Operating current	40.1 mA
Slope efficiency	0.0625

Table 3
EDFA gain simulation parameters [24].

Parameter name	Value
EDFA length	5 m
Output gain at normal condition (G_{In})	17.6 dB
Signal emission cross-section (σ_{se})	$3.41 \times 10^{-25} \text{ m}^2$
Signal absorption cross-section (σ_{sa})	$2.57 \times 10^{-25} \text{ m}^2$
Pump absorption cross-section (σ_{pa})	$2.17 \times 10^{-25} \text{ m}^2$
Er^{+} ion concentration (N_0)	$6.8 \times 10^{24} \text{ m}^{-3}$
Pump wavelength	980 nm
Signal wavelength	1550 nm
Fiber characteristic parameter for pump (Ω_p)	1.949 μm
Fiber characteristic parameter for signal (Ω_s)	1.345 μm

demultiplexer, and optical switch (Table 1). Furthermore, we consider that all lightpaths use PM-QPSK modulation format and carry traffic at a symbol rate of 32 GBaud. The noise bandwidth (B_0) value is taken as 0.1 nm (corresponding to 12.5 GHz at 1550 nm). We consider a network scenario wherein one lightpath is provisioned between any two nodes (i.e., source-destination node pair) in the network. Thus, we consider that in the 28-node network, altogether 756 number of lightpaths exist. The lightpaths are considered to be set up over the shortest path between the source and destination nodes. QoT parameters of all (i.e., 756) lightpaths are recorded at an interval of 10 minutes. We consider a commercially available DWDM transceiver wherein the optical module consists of a tunable external cavity laser (ECL) [32]. The laser currents (operating and threshold) and other parameter values for the ECL during normal operating conditions are shown in Table 2 [32]. To model the gain of an EDFA, we use the simulation parameters shown in Table 3, Eqs. (3), and (4).

4.2. Dataset generation

The given SDON-based network, as illustrated in Fig. 1, has 756 distinct lightpaths. To generate a training dataset for the deep learning models, we record the QoT parameter values for 900 time sample points (which correspond to 150 hours) for each lightpath and each label (i.e., each type of soft failure/no failure scenario). Labels can be of four types, viz., no failure, ECL-, EDFA-, and NLI-related soft failures. Each time sample point (Table 4) consists of five features, namely, lightpath length (km), ECL operating current (mA), received optical power (dBm), OSNR (dB), and BER (dB). In Table 4, we show a snapshot of the training dataset for an example lighpath for no failure and ECL failure scenarios. Under normal conditions, all 756 lightpaths have the transmit optical power following a Gaussian distribution of

Table 4
Dataset for QoT features of lightpaths under normal and ECL failure scenarios.

Time stamp	Lightpath length (km)	ECL current (mA)	Lightpath power (dBm)	OSNR (dB)	BER (dB)	Failure type
1	514	40.1094	-2.0011	24.7464	-267.3813	No failure
2	514	40.0967	-2.0057	24.7441	-267.2442	No failure
⋮	⋮	⋮	⋮	⋮	⋮	⋮
900	514	40.0839	-1.9923	24.7508	-267.6393	No failure
1	514	41.4546	-2.0020	24.7460	-267.3556	ECL failure
2	514	41.4213	-2.0003	24.7468	-267.4029	ECL failure
⋮	⋮	⋮	⋮	⋮	⋮	⋮
900	514	76.1579	-2.0044	24.7447	-267.2823	ECL failure

Table 5
ECL failure simulation parameters.

Parameters	Mean	Standard Deviation
q	1.2	0.05
E_a	0.42 eV	0.01 eV
μ_0	5	0.25
β	1.26 mA	0.05 mA

mean -2 dBm and standard deviation of -30 dBm. The mean value of the operating current of the ECL is 40.1 mA, and all EDFAs have a noise figure of 5 dB. We generate ECL-related soft failures due to increase in threshold current (considering that ECL operates in constant power mode) using Eq. (2). The derating exponent (q), activation energy (E_a), scale parameter (μ_0), and nonradiative current (β) values are considered to follow Gaussian distributions, with the mean and standard deviation shown in Table 5. We introduce additional randomness by adding a Gaussian noise term to the ECL operating current. To induce EDFA gain-related failures for inline amplifiers, we modify the characteristics of an inline EDFA by decreasing its pump laser power and thereby reducing its gain. Any reduction in the EDFA gain of an inline amplifier causes a reduction in the received power of the lightpaths that pass through the amplifier. We employ Eq. (3) and Eq. (4) to alter the gain of the inline EDFA by decreasing the pump laser power. EDFA gain-related failures are generated by randomly decreasing the pump laser power from 5.1 mW with the multiples of step size of 10 μ W. It may be observed from Eq. (9) that the increase in the transmit optical power of a lightpath increases the NLI power at the end of the lightpath. To create NLI-related failure scenarios, we randomly increase the lightpath transmit power from 2 dBm with multiples of the step size of 0.02 dBm.

The training dataset contains 900 time sample points (representing data for 150 hours, recorded at an interval of 10 minutes) for each lightpath and each no failure/failure scenario. Therefore, for 756 lightpaths and four labels, we have a total of 2721600 ($= 900 \times 756 \times 4$) time sample points. We divide the time sample points into two parts: normally operating lightpaths' time sample points and failure lightpaths' time sample points. Normally operating lightpaths' time sample points are used for training the LSTM-AE model, whereas the failure lightpaths' time sample points are used for training the LSTM-classifier model. Furthermore, the test dataset has 300 time sample points (representing data for 50 hours, recorded at an interval of 10 minutes) for a given lightpath and a given label (i.e., no failure/failure type). Our test dataset contains 907200 ($= 300 \times 756 \times 4$) time sample points considering all 756 lightpaths and four types of label. For training and testing purposes of the deep learning models, we create sequences of 10 consecutive time sample points (by sliding window with one time sample point at a time) for a given lightpath and a given label. Thus, two consecutive sequences contain 9 common time sample points. During generation of consecutive sequences, we discard the first-time sample point, which may not be very reliable since OPM data is selected for the first time. We obtain 672840 ($= 890 \times 756 \times 1$) and 2018520 ($= 890 \times 756 \times 3$) sequences for the training (along with validation) of the

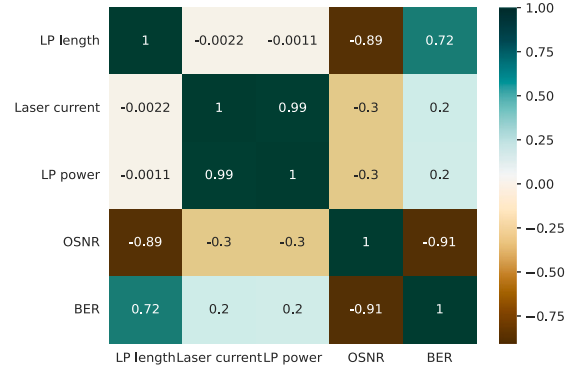


Fig. 5. Correlation matrix.

LSTM-AE and LSTM-classifier, respectively. Our test dataset contains 876960 ($= 290 \times 756 \times 4$) sequences.

Fig. 5 shows the correlation matrix for the training dataset's features, providing a numerical representation of the degree to which pairs of variables are linearly related to each other. Each element of the matrix is the correlation coefficient between the two features, and its value ranges between -1 and $+1$. Fig. 6 shows the variation of different QoT values with time for an example lightpath of length 3854 km. The 'blue' lines in all the subfigures represent the lightpath's normal operation (i.e., not suffering from any type of soft failure). The 'red', 'green', and 'black' lines in all subfigures represent the ECL-, EDFA-, and NLI-related soft failures, respectively. Fig. 6(a) shows the variation of ECL current for different soft failures and no failure conditions. During ECL failure, the operating current increases rapidly to maintain the constant lightpath power. During NLI-related failure, the received optical power increases gradually; moreover, the ECL operating current also increases. During normal operation and EDFA-related failure, the variation of laser current remains the same. Fig. 6(b) shows the received optical power variation with time for different soft failures and no failure conditions. During normal conditions and ECL failure, the received optical power level remains constant, whereas during NLI failure, the power level gradually increases. Moreover, during EDFA failure, the power level decreases rapidly. The variation of OSNR and BER with time are shown in Figs. 6(c) and 6(d), respectively. In normal and ECL failure scenarios, OSNR and BER remain constant with time, while in EDFA- and NLI-failure cases, OSNR decreases and BER increases.

5. Results

In this section, we discuss the performance metrics to evaluate the performance of our proposed LSTM-AE- and LSTM classifier-based cascaded deep learning model. Next, we describe the architecture of our proposed model, and finally, we present the numerical results.

5.1. Performance evaluation metrics

A confusion matrix serves as a tabular representation employed to assess the efficacy of a ML model by contrasting the model's predictions against the factual class labels of a given dataset. In the context of the given multiclass classification problem, we construct a standard confusion matrix, wherein each row corresponds to the real or true class and each column corresponds to the predicted class. The numerical entries within the matrix correspond to the frequencies or counts of occurrences that are classified into distinct groups. True positives (TPs) and true negatives (TNs) are the elements correctly classified by the model, and they are placed on the main diagonal of the confusion matrix. False positives (FPs) are the test results which are wrongly classified as positive. False negatives (FNs) are the test results that are incorrectly classified as negative. The *precision* of a particular class is

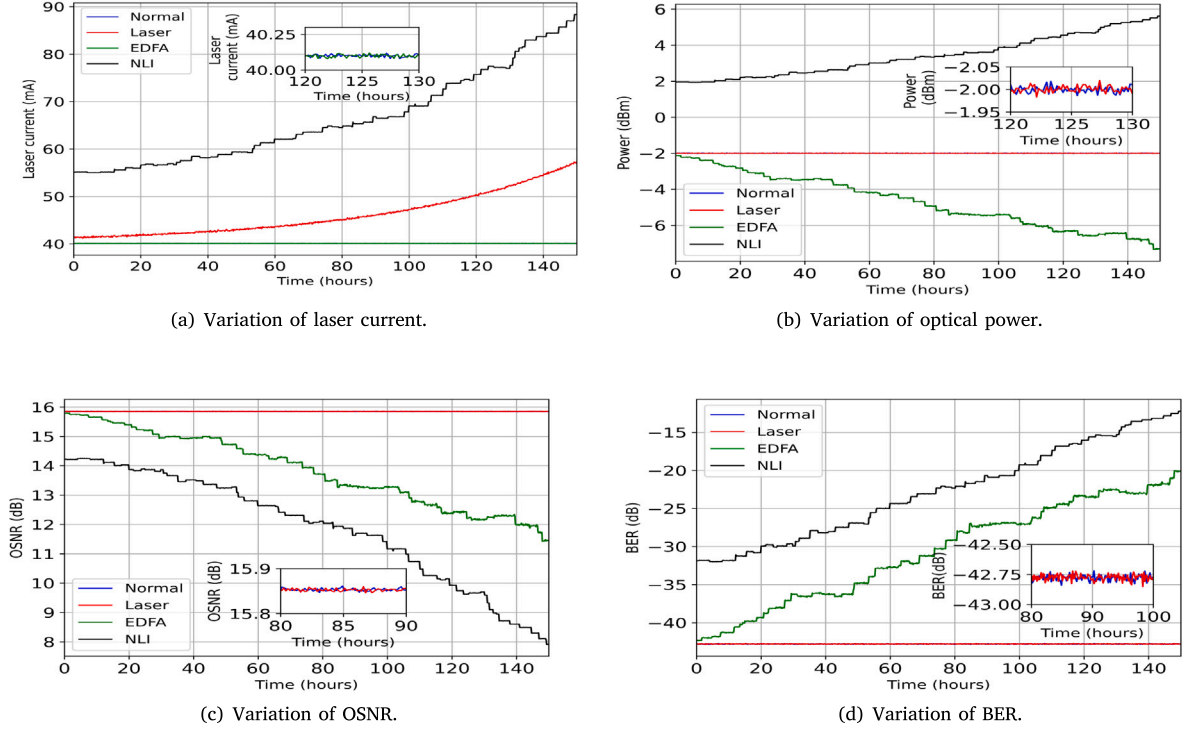


Fig. 6. Variation of QoT values with time for different soft failure/no failure scenarios for an example lightpath of length 3854 km. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

defined as the fraction of TP to the total number of positive predictions, which is given as [33]

$$Precision = \frac{TP}{TP + FP}. \quad (19)$$

The *recall* or the true positive rate (*TPR*) is a measure of the prediction accuracy for a positive class. *TPR* is defined as

$$Recall = \frac{TP}{TP + FN}. \quad (20)$$

The false positive rate (*FPR*) is the ratio of false positive predictions to the total number of actual negative instances. *FPR* is presented as

$$FPR = \frac{FP}{FP + TN}. \quad (21)$$

The *F1-score* is computed as the harmonic mean of precision and recall, and defined as

$$F1 - Score = 2 \frac{Precision \times Recall}{Precision + Recall}. \quad (22)$$

The *F1-score* provides a measure to balance precision and recall. A high value indicates both high precision and high recall for a given class. The accuracy for the multiclass classification is given as

$$Accuracy = \frac{\sum_i TP_i}{\sum_i (TP_i + FP_i + FN_i)}, \quad (23)$$

where TP_i is the number of correctly classified samples for class i , and the denominator represents the total number of instances in the dataset.

5.2. LSTM-AE anomaly detection model

In our study, we employ two deep learning models for soft failure detection and identification. In the first stage, we use an LSTM-AE model as a binary classifier to identify aberrant time series sequences. Subsequently, a multiclass LSTM-based classifier is employed to identify the type of soft failure.

In the training phase of the LSTM-AE model, we utilize only the normal state sequences. However, in the testing phase, both the regular and faulty (i.e., anomalous) data sequences are employed. The encoder component of the proposed AE model comprises two stacked LSTM layers, the first containing 64 LSTM cells and the second having 32 LSTM cells. The input to the first LSTM layer of the encoder is a 2D array consisting of 10 time steps and 5 features representing a 10×5 array. The second LSTM layer has the input size of 5×64 . The output from the second LSTM layer is a 1×32 encoded feature vector (i.e., the latent space). The 'RepeatVector' layer is an intermediary component connecting the encoder and the decoder modules. 'RepeatVector' is used to duplicate the encoded feature vector 10 times. The 2D input array is fed to the first LSTM layer in the decoder. Since the decoder is specifically intended to reverse the encoding process, the LSTM layers in the decoder are arranged in a reversed sequence compared to the encoder. The decoder comprises two LSTM layers. The initial layer comprises 32 LSTM cells, while the subsequent layer comprises 64 LSTM cells. The output from the second LSTM layer of the decoder is a 10×64 array. Thereafter, a 'TimeDistributed' dense layer is added to obtain the reconstructed sequence. The 'TimeDistributed' layer creates a vector of length equal to the number of features (i.e., 64) outputted from the second LSTM layer of the decoder. The 'TimeDistributed' layer creates a long vector and duplicates it 5 (i.e., the number of features in the original dataset) times. The output from the LSTM layer and 'TimeDistributed' layer are 10×64 array and 64×5 array, respectively. A matrix multiplication between the arrays yields a 10×5 output. Thus, the model is trained to determine a threshold value of reconstruction error. During the testing phase, the sequences with reconstruction error values above the threshold are identified as anomalous. We train and validate the model using 70% and 30% of 672840 total data sequences, respectively. We apply a standard scalar normalization technique to remove the impact of different scales on QoT data sequences. We conduct exhaustive trials to obtain the best model by adjusting different LSTM-AE model parameters. In Table 6, we show the best model obtained by utilizing the 'Adam' optimizer (known for its good performance on

Table 6

Hyperparameters for training deep learning models.

Hyperparameter	LSTM-AE	LSTM classifier
Input timesteps	10	10
Learning rate	0.00001	0.000006
Batch size	256	512
Optimizer	Adam	Adam
Epochs	70	40

Table 7

Confusion matrix for the LSTM-AE model.

		Predicted class	
		Class	
Actual class	Normal	219 240	0
	Anomaly	1343	656 377

large datasets) with sequences of 10 timestep length, batch size of 256, and a learning rate (LR) of 0.00001.

5.3. LSTM classifier model

LSTM classifier aims to classify soft failures attributed to ECL, EDFA, or NLI, after a multivariate QoT sequence is detected as an anomaly. We propose an LSTM classifier model consisting of 4 hidden layers. The first three hidden layers are made of LSTM cells, each with 32 neurons, which capture the temporal dependencies in the input sequence. After the third LSTM layer, we use a dense layer with 32 neurons that can learn complex patterns in the input data. Finally, we use an output layer with 4 neurons. We use 'ReLU' activation function for the dense layer and the softmax activation function for the output layer. We use 70% and 30% of 2018520 total data sequences for training and validation, respectively. The proposed model can classify the input sequences into one of the three classes based on the learned patterns within the data. Table 6 shows the parameters for the best LSTM classifier model obtained after exhaustive experimental studies. The model uses the 'Adam' optimizer. We use the categorical cross-entropy loss function, and the batch size is 512. The proposed LSTM-AE and LSTM classifier models are implemented in Keras, which is a powerful and popular deep learning framework that allows efficient and scalable training on large datasets.

5.4. Performance of proposed cascaded deep learning model

In this subsection, first, we discuss the performance of the proposed LSTM-AE model. Next, we discuss the performance of the proposed LSTM classifier model, and finally, the performance of our proposed cascaded model is evaluated. Fig. 7 shows the train/validation loss curves for different LRs of 0.001, 0.0001, and 0.00001, with the batch size and input timestep fixed at 256 and 10 respectively. The curves with 'blue', 'green', and 'red' colors show the loss curves for LRs 0.00001, 0.0001, and 0.001, respectively. With LRs of 0.001 and 0.0001, the loss initially decreases rapidly but then oscillates, indicating potential instability or overfitting. With an LR of 0.00001, the training loss decreases steadily and converges smoothly, with the validation loss showing a similar pattern, suggesting a good balance between training and generalization. We obtain the threshold value of the reconstruction error to be 1.023. Any QoT sequence having a reconstruction error above this threshold is considered to be anomalous. Table 7 shows the performance of the LSTM-AE model based on the confusion matrix. The model is tested over 876960 data sequences. All normal sequences are correctly identified and 1343 anomaly sequences are identified as normal. LSTM-AE has an accuracy of 99.85%.

After detecting anomalous QoT sequences, we use the LSTM classifier to identify the type of soft failure. The classifier classifies the

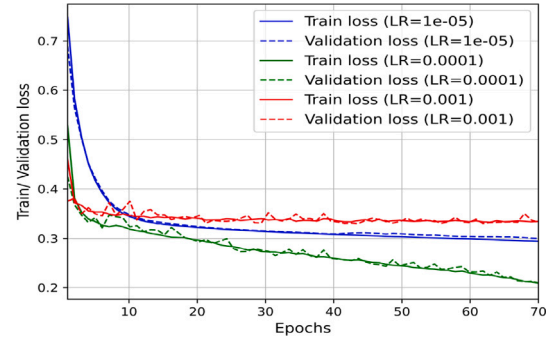


Fig. 7. Loss vs. no. of epochs for the LSTM-AE model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

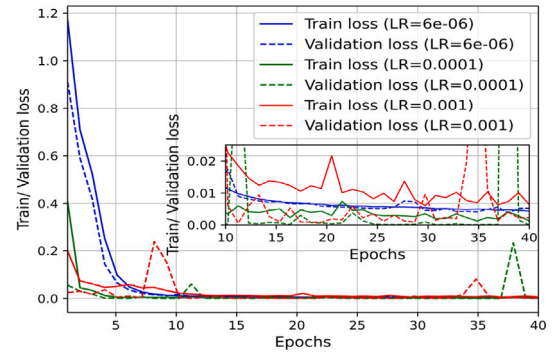


Fig. 8. Loss vs. no. of epochs for the LSTM classifier model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

anomaly sequences into three types: ECL-, EDFA-, and NLI-related soft failures. In our case, 656377 data sequences are identified as anomaly sequences and these sequences are passed through the LSTM classifier. Fig. 8 shows the training and validation loss curves of the LSTM classifier model for different LRs of 0.0001, 0.00001, and 0.000006, keeping the batch size and the input timestep fixed at 512 and 10, respectively. The curves with 'red', 'green', and 'blue' colors show the loss curves for LRs 0.001, 0.0001, and 0.000006, respectively. With an LR of 0.000006, the train/validation loss exhibits a consistent and smooth reduction, indicating a good balance between training and generalization. Table 8 shows the confusion matrix for the proposed LSTM-AE and LSTM classifier-based cascaded soft failure detection and identification model. No failure, ECL-, EDFA-, and NLI-related failures are represented as the class '0', '1', '2', and '3', respectively. Each class has an equal number (viz., 219240) of test data sequences. The model performs exceptionally well for classes '0' and '1', indicating high precision and recall. Although the performance is generally good for class '3', there are 1035 misclassifications into class '1', suggesting some overlap between these classes. Class '2' shows the misclassification, particularly with class '0' (1343 instances) and some with class '1' (266 instances). This indicates a moderate level of class overlap.

Using the confusion matrix, key performance metrics, such as precision, recall, and F1-score are calculated for each class (as shown in Table 9). Notably, classes '0' and '1' have perfect recall (100%) indicating that the model does not overlook any instances of these classes. Recall for classes '2' and '3' is marginally lower (99.27% and 99.53%), suggesting a tiny percentage of cases that are incorrectly classified. However, their precisions stay at 100%, guaranteeing that

Table 8

Confusion matrix for the proposed cascaded model.

Actual class	Predicted class				
	Class	0	1	2	3
	0	219 240	0	0	0
	1	0	219 235	5	0
	2	1343	266	217 631	0
	3	0	1035	0	218 205

Table 9

Summary of results for the proposed cascaded model.

Class	Precision (%)	Recall (%)	F1-score (%)
0	99.39	100	99.69
1	99.41	100	99.70
2	100	99.27	99.63
3	100	99.53	99.76

Table 10

Accuracy of proposed cascaded model for different timesteps.

Time steps	Total test sequence	LSTM-AE accuracy (%)	Total test anomaly sequence	LSTM classifier accuracy (%)	Cascaded model accuracy (%)
6	889056	99.79	666792	99.43	99.45
8	883008	99.80	662256	99.80	99.65
10	876960	99.85	657720	99.80	99.70
12	870912	99.89	653184	99.33	99.39

Table 11

Performance comparison of proposed cascaded model with other ML models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Proposed model	99.70	99.70	99.70	99.70
LSTM	98.26	98.37	98.26	98.26
GRU	95.78	96.38	95.78	95.75
RF	96.99	97.31	96.99	96.98
SVM	85.44	89.11	85.44	84.52

the model makes accurate predictions for the classes. Our cascaded model shows an accuracy of 99.70% (Table 10). In Table 10, we also show the accuracy of our cascaded model with different timesteps, keeping the batch size and the learning rate fixed at 512 and 0.000006, respectively. First, we calculate the accuracy of LSTM-AE model over the test sequences (consisting of normal and anomalous sequences). Then, we calculate the accuracy of the LSTM classifier over the test anomaly sequences. Finally, the accuracy of the cascaded model is evaluated over the test sequences. We obtain the best performance with 10 timesteps. Table 11 shows the comparison in performance of our proposed model with several ML-based classifiers, developed using LSTM, GRU, RF, and SVM, which classify data sequences into four classes, viz., no failure, ECL-, EDFA-, and NLI-related failures in a single stage only. Our proposed model outperforms the other ML models in terms of accuracy, precision, recall, and F1-score.

6. Conclusion

We proposed a cascaded soft failure detection and identification framework for a backbone optical network using LSTM-based deep learning models. Our proposed cascaded model almost accurately detects and identifies ECL-, EDFA-, and NLI-related soft failures. The

confusion matrix for the proposed model reveals minimal misclassifications, highlighting the model's robustness and accuracy. The model demonstrates exceptional performance across all classes, with near-perfect precision, recall, and F1 scores, over classifiers developed using LSTM, GRU, RF, and SVM. Overall, the high accuracy and balanced performance underscore the effectiveness of the cascaded model in multiclass classification tasks.

CRedit authorship contribution statement

Subhendu Ghosh: Writing – original draft, Validation, Software, Resources, Methodology, Formal analysis, Data curation. **Aneek Adhya:** Writing – review & editing, Visualization, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Training and testing soft failure dataset is available at : <https://data.mendeley.com/datasets/y3pspy7j83/1>.

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